

## TOLI-TOLI LALOS, Indonesia

WMO: 970280

Lat: 1.1211N

Lon: 120.7942E

Elev: 7

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 00-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |       |      |      |       |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-------|------|------|-------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |       |      | 99%  |       |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR    | MCDB | DP   | HR    | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)   | (f)  | (g)  | (h)   | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 72.5       | 73.1 | 68.9                          | 106.1 | 76.6 | 70.0 | 110.3 | 76.5 | 12.9                  | 82.6 | 9.9 | 84.6 | 2.2                   | 110  | 0.290 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 5             | 12.7                      | 90.6            | 79.4 | 89.8 | 79.4 | 89.1 | 79.3 | 81.2                | 87.8 | 80.8 | 87.6 | 80.5 | 87.2 | 5.3                     | 320  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 79.2                            | 151.7 | 85.4 | 78.9 | 149.9 | 85.3 | 78.3 | 147.1 | 84.9 | 44.9          | 88.2 | 44.5 | 88.0 | 44.1 | 87.7 | 85.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|-----|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |     | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 9.9           | 8.7  | 7.4 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |     | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 81.1   | 80.4  | 80.5  | 80.8  | 81.3  | 81.6  | 81.2  | 81.0  | 81.1  | 81.3  | 81.2  | 81.2  | 81.1  |
|                                                                        | DBStd       | 1.41   | 1.58  | 1.51  | 1.40  | 1.13  | 1.24  | 1.32  | 1.46  | 1.51  | 1.25  | 1.22  | 1.12  | 1.53  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | CDD50       | 11334  | 941   | 855   | 955   | 939   | 981   | 935   | 960   | 963   | 939   | 967   | 936   | 963   |
|                                                                        | CDD65       | 5859   | 476   | 435   | 490   | 489   | 516   | 485   | 495   | 498   | 489   | 502   | 486   | 498   |
|                                                                        | CDH74       | 57772  | 4371  | 4034  | 4710  | 5025  | 5421  | 4740  | 4855  | 4957  | 4958  | 4977  | 4804  | 4920  |
| (13)                                                                   | CDH80       | 19871  | 1248  | 1212  | 1546  | 1810  | 1975  | 1632  | 1683  | 1815  | 1859  | 1812  | 1675  | 1604  |
| (14)                                                                   | Wind        | WSAvg  | 2.7   | 2.6   | 2.5   | 2.5   | 2.5   | 2.4   | 2.5   | 2.5   | 2.9   | 3.1   | 2.9   | 2.8   |
| (15) Precipitation                                                     | PrecAvg     | 82.8   | 9.2   | 6.5   | 5.6   | 4.9   | 6.2   | 7.0   | 7.8   | 5.6   | 4.9   | 7.0   | 6.7   | 8.2   |
|                                                                        | PrecMax     | 146.1  | 24.1  | 16.9  | 14.6  | 11.6  | 13.9  | 17.0  | 21.9  | 14.5  | 14.0  | 20.8  | 17.6  | 17.9  |
|                                                                        | PrecMin     | 44.7   | 1.3   | 0.0   | 0.1   | 0.0   | 1.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 1.4   | 1.8   |
|                                                                        | PrecStd     | 24.8   | 5.4   | 4.6   | 4.1   | 2.7   | 2.8   | 4.6   | 5.0   | 4.4   | 4.3   | 5.4   | 3.6   | 4.6   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 89.7  | 89.9  | 90.9  | 90.8  | 90.7  | 90.1  | 90.2  | 90.7  | 90.7  | 91.0  | 90.5  | 90.6  |
|                                                                        |             | MCWB   | 79.4  | 79.5  | 79.3  | 79.7  | 80.1  | 79.8  | 79.1  | 78.7  | 78.7  | 79.0  | 79.3  | 79.7  |
|                                                                        | 2%          | DB     | 88.1  | 88.2  | 89.0  | 89.1  | 89.4  | 89.0  | 89.1  | 89.5  | 89.5  | 89.5  | 89.1  | 89.1  |
|                                                                        |             | MCWB   | 78.8  | 78.8  | 79.1  | 79.6  | 80.0  | 79.8  | 79.2  | 79.0  | 79.1  | 79.2  | 79.4  | 79.2  |
|                                                                        | 5%          | DB     | 87.1  | 87.2  | 87.9  | 88.2  | 88.6  | 88.1  | 88.2  | 88.6  | 88.7  | 88.6  | 88.1  | 88.0  |
|                                                                        |             | MCWB   | 78.5  | 78.5  | 78.8  | 79.4  | 79.9  | 79.6  | 79.1  | 78.9  | 79.0  | 79.2  | 79.3  | 79.1  |
|                                                                        | 10%         | DB     | 86.0  | 86.1  | 86.8  | 87.4  | 87.8  | 87.2  | 87.3  | 87.7  | 87.8  | 87.6  | 87.2  | 86.9  |
|                                                                        |             | MCWB   | 78.3  | 78.3  | 78.5  | 79.2  | 79.8  | 79.3  | 78.9  | 78.8  | 78.9  | 79.1  | 79.1  | 78.8  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 80.7  | 81.3  | 80.9  | 81.2  | 81.7  | 81.4  | 81.0  | 81.0  | 81.2  | 81.0  | 81.0  | 81.0  |
|                                                                        |             | MCDB   | 87.0  | 88.2  | 87.8  | 88.0  | 87.8  | 88.1  | 87.7  | 88.1  | 88.2  | 87.9  | 87.1  | 87.8  |
|                                                                        | 2%          | WB     | 79.8  | 79.9  | 80.0  | 80.5  | 80.9  | 80.7  | 80.4  | 80.4  | 80.6  | 80.4  | 80.4  | 80.2  |
|                                                                        |             | MCDB   | 86.1  | 86.3  | 86.9  | 87.5  | 87.3  | 87.4  | 87.2  | 87.5  | 87.4  | 87.3  | 87.1  | 86.8  |
|                                                                        | 5%          | WB     | 79.2  | 79.2  | 79.5  | 80.0  | 80.5  | 80.0  | 79.9  | 79.8  | 80.0  | 79.9  | 79.9  | 79.7  |
|                                                                        |             | MCDB   | 85.3  | 85.4  | 86.3  | 86.8  | 87.0  | 86.7  | 86.6  | 86.9  | 86.9  | 86.8  | 86.5  | 86.3  |
|                                                                        | 10%         | WB     | 78.7  | 78.6  | 78.9  | 79.5  | 80.0  | 79.6  | 79.3  | 79.2  | 79.4  | 79.4  | 79.3  | 79.1  |
|                                                                        |             | MCDB   | 84.6  | 84.7  | 85.5  | 86.2  | 86.3  | 86.1  | 85.9  | 86.2  | 86.4  | 86.2  | 85.9  | 85.5  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.0  | 11.6  | 12.5  | 13.0  | 12.7  | 12.4  | 12.7  | 13.5  | 13.6  | 13.4  | 12.8  | 11.9  |
|                                                                        |             | MCDBR  | 12.8  | 13.2  | 14.2  | 14.1  | 13.8  | 13.3  | 13.9  | 14.6  | 14.7  | 14.7  | 14.1  | 13.4  |
|                                                                        |             | MCWBR  | 5.9   | 6.2   | 6.7   | 6.5   | 6.5   | 6.5   | 6.7   | 6.9   | 7.1   | 7.0   | 6.9   | 6.3   |
|                                                                        | 5% WB       | MCDBR  | 11.9  | 12.2  | 13.1  | 13.3  | 13.0  | 12.7  | 13.1  | 13.8  | 13.7  | 13.8  | 13.2  | 12.6  |
|                                                                        |             | MCWBR  | 6.0   | 6.1   | 6.6   | 6.6   | 6.5   | 6.6   | 6.8   | 7.0   | 7.2   | 7.0   | 6.8   | 6.4   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.405 | 0.409 | 0.411 | 0.409 | 0.404 | 0.405 | 0.404 | 0.431 | 0.460 | 0.502 | 0.429 | 0.409 |
|                                                                        | taud        |        | 2.523 | 2.503 | 2.494 | 2.498 | 2.523 | 2.536 | 2.540 | 2.439 | 2.348 | 2.212 | 2.436 | 2.512 |
|                                                                        | Ebn at Noon |        | 293   | 292   | 290   | 284   | 279   | 274   | 276   | 274   | 272   | 263   | 283   | 290   |
|                                                                        | Edh at Noon |        | 35    | 36    | 36    | 35    | 33    | 32    | 32    | 37    | 41    | 47    | 37    | 35    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1513  | 1661  | 1788  | 1797  | 1718  | 1585  | 1642  | 1767  | 1839  | 1739  | 1659  | 1539  |
|                                                                        | RadStd      |        | 112   | 135   | 154   | 117   | 77    | 86    | 130   | 94    | 117   | 109   | 92    | 122   |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page